

Problem

One line voltage of a balanced Y-connected source is $\mathbf{V}_{AB} = 120\angle -20^\circ \text{ V}$. If the source is connected to a Δ -connected load of $20\angle 40^\circ \Omega$, find the phase and line currents. Assume the abc sequence.

Step-by-step solution

Step 1 of 6

Draw the circuit as shown in Figure 1.

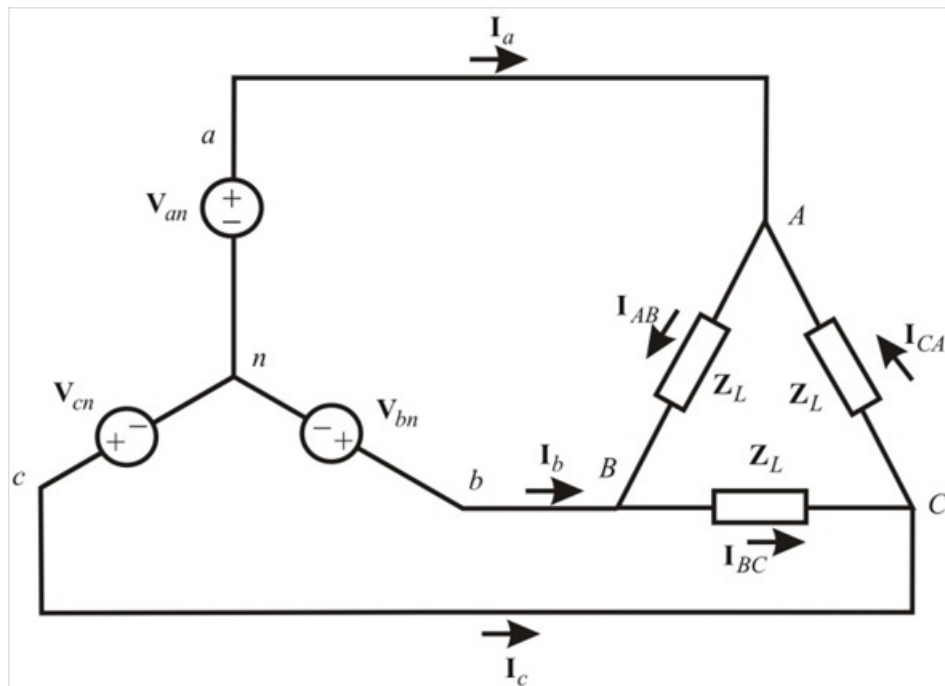


Figure 1

Step 2 of 6

Consider line voltage of the star connected load as,

$$V_{AB} = 120\angle -20^\circ \text{ V}$$

Consider the load impedance of the load is,

$$Z_{\Delta} = 20\angle 40^\circ \Omega$$

Determine the phase current between A and B .

Consider the equation for line current I_{AB} .

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}} \dots\dots (1)$$

Replace $120\angle -120^\circ$ for V_{AB} , and $20\angle 40^\circ$ for Z_{Δ} in equation 1.

$$\begin{aligned} I_{AB} &= \frac{120\angle -120^\circ}{20\angle 40^\circ} \\ &= 6\angle -60^\circ \text{ A} \end{aligned}$$

Therefore, the line current I_{AB} is $6\angle -60^\circ$.

Step 3 of 6

Determine the phase current between B and C .

Consider,

$$\begin{aligned} I_{BC} &= I_{AB}\angle -120^\circ \\ I_{BC} &= 6\angle (-60^\circ - 120^\circ) \\ &= 6\angle -180^\circ \text{ A} \end{aligned}$$

Determine the phase current between C and A .

Consider,

$$\begin{aligned} I_{CA} &= I_{AB}\angle 120^\circ \\ I_{BC} &= 6\angle (-60^\circ + 120^\circ) \\ &= 6\angle 60^\circ \text{ A} \end{aligned}$$

Therefore, the line current I_{BC} is $6\angle -180^\circ$.

Therefore, the line current I_{CA} is $6\angle 60^\circ$.

Step 4 of 6

Determine the line current at a .

Consider the equation for line current.

$$I_a = I_{AB}\sqrt{3}\angle -30^\circ$$

Replace $6\angle -60^\circ \text{ A}$ for I_{AB} in equation.

$$\begin{aligned} I_a &= 6 \times \sqrt{3}\angle (-60^\circ - 30^\circ) \\ &= 10.392\angle -90^\circ \text{ A} \end{aligned}$$

Therefore, the line current I_a is $10.392\angle -90^\circ \text{ A}$.

Step 5 of 6

Determine the line current at b .

Consider the equation for line current.

$$\mathbf{I}_b = \mathbf{I}_a \angle -120^\circ$$

Replace $10.392 \angle -90^\circ \text{ A}$ for $\mathbf{I}_a$ in equation.

$$\begin{aligned}\mathbf{I}_b &= (10.392 \angle -90^\circ \text{ A})(\angle -120^\circ) \\ &= -8.997 + j5.196 \\ &= 10.392 \angle 150^\circ\end{aligned}$$

Therefore, the line current $\mathbf{I}_b$ is $10.392 \angle 150^\circ \text{ A}$.

Step 6 of 6

Determine the line current at c .

Consider the equation for line current.

$$\mathbf{I}_c = \mathbf{I}_a \angle 120^\circ$$

Replace $10.392 \angle -90^\circ \text{ A}$ for $\mathbf{I}_a$ in equation.

$$\begin{aligned}\mathbf{I}_c &= 10.392 \angle (-90^\circ + 120^\circ) \text{ A} \\ &= 10.392 \angle 30^\circ \text{ A}\end{aligned}$$

Therefore, the line current $\mathbf{I}_c$ is $10.392 \angle -210^\circ \text{ A}$.